

HOW ACCURATE IS GOOD ENOUGH?

Approximating a Civil Engineering Function Using Infinite Series — Numerical Tables (Deliverable 1)

[1. Numerical Tables](#) [2. Written Recommendation](#)

1. Numerical Tables

1.1 Geometric Series — $1 + x + x^2 + x^3 + \dots = 1/(1-x)$

Partial sums computed term by term with a loop (no closed-form shortcut). Larger x (closer to 1) converges far more slowly.

x = 0.5 (exact sum = $1/(1-x) = 2.0000000000$)

N (terms)	Partial Sum S ₇	Exact	Abs Error	Rel Error %
5	1.9687500000	2.0000000000	3.12e-02	1.56250000
10	1.9990234375	2.0000000000	9.77e-04	0.04882812
20	1.999990463	2.0000000000	9.54e-07	0.00004768
50	2.0000000000	2.0000000000	8.88e-16	0.00000000
100	2.0000000000	2.0000000000	0.00e+00	0.00000000

x = 0.8 (exact sum = $1/(1-x) = 5.0000000000$)

N (terms)	Partial Sum S ₇	Exact	Abs Error	Rel Error %
5	3.6892800000	5.0000000000	1.31e+00	26.21440000
10	4.5705032704	5.0000000000	4.29e-01	8.58993459
20	4.9538831398	5.0000000000	4.61e-02	0.92233720
50	4.9999429101	5.0000000000	5.71e-05	0.00114180
100	4.999999992	5.0000000000	8.15e-10	0.00000002

x = 0.9 (exact sum = $1/(1-x) = 10.0000000000$)

N (terms)	Partial Sum S ₇	Exact	Abs Error	Rel Error %
5	4.6855900000	10.0000000000	5.31e+00	53.14410000
10	6.8618940391	10.0000000000	3.14e+00	31.38105961
20	8.9058101087	10.0000000000	1.09e+00	10.94189891
50	9.9536160231	10.0000000000	4.64e-02	0.46383977
100	9.9997609474	10.0000000000	2.39e-04	0.00239053

1.2 Power Series — $P_h(x) = a_0 + a_1x + a_2x^2 + \dots + a_hx^N$

A polynomial is a finite power series; an infinite power series can approximate transcendental functions.

$P_h(x)$ for $1/(1-x)$ evaluated at x = 0.5 (coefficients $a_s = 1$)

N	Power Series Value	Exact $1/(1-x)$	Abs Error
5	1.9687500000	2.0000000000	3.12e-02
10	1.9990234375	2.0000000000	9.77e-04
20	1.999990463	2.0000000000	9.54e-07

1.3 Maclaurin Series for sin(θ) — θ = 10°

$\sin(\theta) = \theta - \theta^3/3! + \theta^5/5! - \theta^7/7! + \dots$ (multiplication to $\theta^{15} = 8$ terms is already exact to machine precision)

Maclaurin approximation of sin(10°) — exact value = 0.173648177667

Terms	Approximation	Exact sin(10°)	Abs Error	Rel Error %
1	0.174532925199	0.173648177667	8.85e-04	0.50950580
2	0.173646829044	0.173648177667	1.35e-06	0.00077664
3	0.173648178645	0.173648177667	9.78e-10	0.00000056
4	0.173648177667	0.173648177667	4.14e-13	0.00000000
5	0.173648177667	0.173648177667	1.11e-16	0.00000000
6	0.173648177667	0.173648177667	0.00e+00	0.00000000
7	0.173648177667	0.173648177667	0.00e+00	0.00000000
8	0.173648177667	0.173648177667	0.00e+00	0.00000000

1.4 Engineering Investigation — $y = 20 \cdot \sin(\theta)$

Vertical component $y = L \cdot \sin(\theta)$, $L = 20$ m, angles 1°, 2°, 5°, 10°, 15°, 20°, 30°. Error grows with angle and shrinks as terms are added.

Maclaurin series — 1 term(s)

Angle	Exact y (m)	Approx y (m)	Abs Error (m)	% Error
1°	0.34904813	0.34906585	0.000018	0.005077
2°	0.69798993	0.69813170	0.000142	0.020311
5°	1.74311485	1.74532925	0.002214	0.127037
10°	3.47296355	3.49065850	0.017695	0.509506
15°	5.17638090	5.23598776	0.059607	1.151516
20°	6.84040287	6.98131701	0.140914	2.060027
30°	10.00000000	10.47197551	0.471976	4.719755

Maclaurin series — 2 term(s)

Angle	Exact y (m)	Approx y (m)	Abs Error (m)	% Error
1°	0.34904813	0.34904813	0.000000	0.000000
2°	0.69798993	0.69798993	0.000000	0.000001
5°	1.74311485	1.74311401	0.000001	0.000048
10°	3.47296355	3.47293658	0.000027	0.000777
15°	5.17638090	5.17617627	0.000205	0.003953
20°	6.84040287	6.83954162	0.000861	0.012591
30°	10.00000000	9.99348359	0.006516	0.065164

Maclaurin series — 3 term(s)

Angle	Exact y (m)	Approx y (m)	Abs Error (m)	% Error
1°	0.34904813	0.34904813	0.000000	0.000000
2°	0.69798993	0.69798993	0.000000	0.000000
5°	1.74311485	1.74311486	0.000000	0.000000
10°	3.47296355	3.47296357	0.000000	0.000001
15°	5.17638090	5.17638124	0.000000	0.000006
20°	6.84040287	6.84040537	0.000003	0.000037
30°	10.00000000	10.00004265	0.000043	0.000427

Maclaurin series — 4 term(s)

Angle	Exact y (m)	Approx y (m)	Abs Error (m)	% Error
1°	0.34904813	0.34904813	0.000000	0.000000
2°	0.69798993	0.69798993	0.000000	0.000000
5°	1.74311485	1.74311485	0.000000	0.000000
10°	3.47296355	3.47296355	0.000000	0.000000
15°	5.17638090	5.17638090	0.000000	0.000000
20°	6.84040287	6.84040286	0.000000	0.000000
30°	10.00000000	9.99999984	0.000000	0.000002

1.5 Taylor Series centered at a = 10° — y = 20·sin(θ)

Taylor generalizes Maclaurin to any center a. Best accuracy is achieved near the expansion point.

Taylor series centered at 10° — 1 term(s)

Angle	Exact y (m)	Approx y (m)	Abs Error (m)	% Error
1°	0.34904813	3.47296355	3.123915	894.981284
2°	0.69798993	3.47296355	2.774974	397.566424
5°	1.74311485	3.47296355	1.729849	99.238940
10°	3.47296355	3.47296355	0.000000	0.000000
15°	5.17638090	3.47296355	1.703417	32.907496
20°	6.84040287	3.47296355	3.367439	49.228669
30°	10.00000000	3.47296355	6.527036	65.270364

Taylor series centered at 10° — 2 term(s)

Angle	Exact y (m)	Approx y (m)	Abs Error (m)	% Error
1°	0.34904813	0.37909875	0.030051	8.609306
2°	0.69798993	0.72286151	0.024872	3.563314
5°	1.74311485	1.75414977	0.011035	0.633058
10°	3.47296355	3.47296355	0.000000	0.000000
15°	5.17638090	5.19177733	0.015396	0.297436
20°	6.84040287	6.91059111	0.070188	1.026083
30°	10.00000000	10.34821867	0.348219	3.482187

Taylor series centered at 10° — 3 term(s)

Angle	Exact y (m)	Approx y (m)	Abs Error (m)	% Error
1°	0.34904813	0.33625278	0.012795	3.665783
2°	0.69798993	0.68900790	0.008982	1.286843
5°	1.74311485	1.74092571	0.002189	0.125588
10°	3.47296355	3.47296355	0.000000	0.000000
15°	5.17638090	5.17855327	0.002172	0.041967
20°	6.84040287	6.85769485	0.017292	0.252792
30°	10.00000000	10.13663363	0.136634	1.366336

Taylor series centered at 10° — 4 term(s)

Angle	Exact y (m)	Approx y (m)	Abs Error (m)	% Error
1°	0.34904813	0.34897579	0.000072	0.020725
2°	0.69798993	0.69794368	0.000046	0.006627
5°	1.74311485	1.74310730	0.000008	0.000434
10°	3.47296355	3.47296355	0.000000	0.000000
15°	5.17638090	5.17637168	0.000009	0.000178
20°	6.84040287	6.84024216	0.000161	0.002349
30°	10.00000000	9.99701213	0.002988	0.029879

1.6 Maclaurin vs Taylor — percentage error comparison

Maclaurin (center 0°) vs Taylor (center 10°) — 1 term(s), relative % error

Angle	Maclaurin % Error	Taylor % Error	Better
1°	0.00507714	894.98128405	Maclaurin
2°	0.02031072	397.56642380	Maclaurin
5°	0.12703678	99.23893962	Maclaurin
10°	0.50950580	0.00000000	Taylor
15°	1.15151599	32.90749620	Maclaurin
20°	2.06002693	49.22866941	Maclaurin

Angle	Maclaurin % Error	Taylor % Error	Better
30°	4.71975512	65.27036447	Maclaurin

Maclaurin (center 0°) vs Taylor (center 10°) — 2 term(s), relative % error

Angle	Maclaurin % Error	Taylor % Error	Better
1°	0.00000008	8.60930630	Maclaurin
2°	0.00000124	3.56331400	Maclaurin
5°	0.00004838	0.63305751	Maclaurin
10°	0.00077664	0.00000000	Taylor
15°	0.00395328	0.29743619	Maclaurin
20°	0.01259054	1.02608349	Maclaurin
30°	0.06516412	3.48218669	Maclaurin

Maclaurin (center 0°) vs Taylor (center 10°) — 3 term(s), relative % error

Angle	Maclaurin % Error	Taylor % Error	Better
1°	0.00000000	3.66578328	Maclaurin
2°	0.00000000	1.28684281	Maclaurin
5°	0.00000001	0.12558814	Maclaurin
10°	0.00000056	0.00000000	Taylor
15°	0.00000646	0.04196687	Maclaurin
20°	0.00003657	0.25279191	Maclaurin
30°	0.00042652	1.36633630	Maclaurin

Maclaurin (center 0°) vs Taylor (center 10°) — 4 term(s), relative % error

Angle	Maclaurin % Error	Taylor % Error	Better
1°	0.00000000	0.02072462	Maclaurin
2°	0.00000000	0.00662723	Maclaurin
5°	0.00000000	0.00043368	Maclaurin
10°	0.00000000	0.00000000	Taylor
15°	0.00000001	0.00017813	Maclaurin
20°	0.00000006	0.00234931	Maclaurin
30°	0.00000163	0.02987869	Maclaurin

1.7 Error Tolerance — minimum terms for < 0.1% error

Minimum number of terms to meet < 0.1% relative error

Angle	Min Terms (Maclaurin)	Min Terms (Taylor@10°)
1°	1	4
2°	1	4
5°	2	4
10°	2	1
15°	2	3
20°	2	4
30°	2	4

1.8 Small-Angle Approximation $\sin(\theta) \approx \theta$

Small-angle approximation $\sin(\theta) \approx \theta$ (1-term)

The $\sin(\theta) \approx \theta$ small-angle approximation stays within the 0.1% tolerance only up to about 4°; it first fails at 5°.

Angle	θ (rad)	$\sin(\theta)$	Abs Error	% Error	OK < 0.1%?
1°	0.017453	0.017452	8.86e-07	0.005077	YES
2°	0.034907	0.034899	7.09e-06	0.020311	YES
3°	0.052360	0.052336	2.39e-05	0.045707	YES

Angle	θ (rad)	$\sin(\theta)$	Abs Error	% Error	OK < 0.1%?
4°	0.069813	0.069756	5.67e-05	0.081278	YES
5°	0.087266	0.087156	1.11e-04	0.127037	NO
6°	0.104720	0.104528	1.91e-04	0.183005	NO
7°	0.122173	0.121869	3.04e-04	0.249205	NO
8°	0.139626	0.139173	4.53e-04	0.325666	NO
9°	0.157080	0.156434	6.45e-04	0.412420	NO
10°	0.174533	0.173648	8.85e-04	0.509506	NO
11°	0.191986	0.190809	1.18e-03	0.616964	NO
12°	0.209440	0.207912	1.53e-03	0.734841	NO
13°	0.226893	0.224951	1.94e-03	0.863187	NO
14°	0.244346	0.241922	2.42e-03	1.002059	NO
15°	0.261799	0.258819	2.98e-03	1.151516	NO
16°	0.279253	0.275637	3.62e-03	1.311624	NO
17°	0.296706	0.292372	4.33e-03	1.482451	NO
18°	0.314159	0.309017	5.14e-03	1.664074	NO
19°	0.331613	0.325568	6.04e-03	1.856571	NO
20°	0.349066	0.342020	7.05e-03	2.060027	NO
21°	0.366519	0.358368	8.15e-03	2.274532	NO
22°	0.383972	0.374607	9.37e-03	2.500181	NO
23°	0.401426	0.390731	1.07e-02	2.737074	NO
24°	0.418879	0.406737	1.21e-02	2.985317	NO
25°	0.436332	0.422618	1.37e-02	3.245021	NO
26°	0.453786	0.438371	1.54e-02	3.516303	NO
27°	0.471239	0.453990	1.72e-02	3.799286	NO
28°	0.488692	0.469472	1.92e-02	4.094098	NO
29°	0.506145	0.484810	2.13e-02	4.400874	NO
30°	0.523599	0.500000	2.36e-02	4.719755	NO
31°	0.541052	0.515038	2.60e-02	5.050887	NO
32°	0.558505	0.529919	2.86e-02	5.394425	NO
33°	0.575959	0.544639	3.13e-02	5.750528	NO
34°	0.593412	0.559193	3.42e-02	6.119363	NO
35°	0.610865	0.573576	3.73e-02	6.501104	NO
36°	0.628319	0.587785	4.05e-02	6.895933	NO
37°	0.645772	0.601815	4.40e-02	7.304038	NO
38°	0.663225	0.615661	4.76e-02	7.725616	NO
39°	0.680678	0.629320	5.14e-02	8.160870	NO
40°	0.698132	0.642788	5.53e-02	8.610012	NO
41°	0.715585	0.656059	5.95e-02	9.073263	NO
42°	0.733038	0.669131	6.39e-02	9.550853	NO
43°	0.750492	0.681998	6.85e-02	10.043018	NO
44°	0.767945	0.694658	7.33e-02	10.550006	NO
45°	0.785398	0.707107	7.83e-02	11.072073	NO